

Roberto André Mossi Milla

Software Engineer | AI Systems | Full-Stack Development

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Summary

Software Engineer building production AI systems and full-stack applications from data processing and model integration through backend architecture and user-facing products. Experience combining LLMs, multimodal AI, computer vision, OCR, cloud services, and modern web technologies to automate workflows and reduce processes that previously required days or weeks to minutes. Published AI researcher with experience spanning evolutionary computation, intelligent systems, and production software engineering.

Experience

Ingeniería Digital

May 2026 – Present

Software Engineer (AI Systems)

Honduras

- Built an AI-assisted medical documentation platform from scratch, reducing post-surgical report preparation from approximately one week to 1 day and completing final document generation in under one minute.
- Architected the full-stack platform using React, Next.js, TypeScript, PostgreSQL, Auth0, and AWS S3, implementing secure document upload, AI processing, editing, versioning, and review workflows.
- Built OCR and language-processing pipelines using Amazon Textract and OpenAI APIs to transform unstructured clinical records into structured, reviewable medical documentation.
- Built a multimodal AI vehicle-damage assessment platform that reduced assessment workflows from more than one week to approximately 1–2 minutes while automatically generating damage analyses and insurance estimates.
- Integrated AWS Rekognition for vehicle localization and Gemini Flash for image-based damage detection, combined with VIN-history retrieval, Playwright data acquisition, PostgreSQL, and AWS S3.
- Developed interactive Konva-based human-validation workflows allowing users to correct AI-generated detection regions and persist annotations for evaluating different models, processing stages, and AI combinations.

Gannon University

Aug. 2024 – May 2025

Computer Vision Engineer — Industry-Sponsored Project

Erie, PA

- Engineered components of a real-time industrial computer-vision system using Ultralytics YOLOv8, LiDAR-triggered image capture, Docker, TimescaleDB, and distributed C++, Python, and Rust services.
- Developed a live monitoring interface and annotation tools allowing operators to review detections, correct bounding boxes, curate training data, and submit revised data for model retraining.
- Developed image-quality and preprocessing workflows using contrast, color, filtering, spatial image partitioning, and reconstruction to improve image usability within the vision pipeline.
- Helped migrate the inference system from Raspberry Pi hardware after thermal and performance limitations were reached, supporting continued real-time inference on NVIDIA Jetson hardware.

Gannon University

Aug. 2024 – May 2025

Research Assistant — Evolutionary Artificial Intelligence

Erie, PA

- Conducted undergraduate research on autonomous multi-agent learning and evolutionary computation using NeuroEvolution of Augmenting Topologies (NEAT), investigating emergent behavior in dynamic environments.
- Independently implemented NEAT in C++ and designed a numerical genome representation encoding agent sensing, actions, and neural-network structure for evolutionary optimization.
- Developed a procedurally generated 3D Unreal Engine simulation environment using C++, and CUDA; research resulted in a peer-reviewed ASEE publication and Sigma Xi presentation at Penn State University.

Dinant

Jun. 2023 – Jul. 2023

Full-Stack Developer

Honduras

- Led end-to-end development of two production platforms used by two companies, replacing manual operational processes that previously required weeks with automated workflows completed in minutes; systems remain in use.
- Designed database-backed workflows for contract management, digital signatures, automated PDF generation, repair tracking, machine maintenance, inventory deduction, and operational reporting.
- Collaborated directly with international stakeholders from requirements gathering through demonstrations, production feedback, and deployment.

Selected Project

NOIR

Jan. 2026 – Present

Founder & Software Engineer

Remote

- Designing and developing a cross-platform movie discovery and continuity platform spanning native Apple applications and a Rust backend built with Axum, PostgreSQL, Redis, authentication, and cloud services.
- Architecting backend services for metadata aggregation, availability tracking, synchronization, and natural-language movie discovery across streaming platforms, theaters, and devices.
- Designing focused AI-assisted movie discovery workflows that translate natural-language viewing intent into structured search and availability queries.

Open Source

Copilot-Lualine

2024 – Present

Creator & Maintainer

Open Source

- Created and maintain an open-source Neovim plugin providing interactive GitHub Copilot status indicators and controls for authentication, enabling/disabling Copilot, and logging; received 117+ GitHub stars and contributions from four community developers.

Technical Skills

Languages: Python, C++, Rust, TypeScript, JavaScript, SQL, Java, Bash, Lua

AI / ML: OpenAI APIs, Gemini Flash, AWS Rekognition, Amazon Textract, Claude APIs, Ultralytics YOLOv8, CUDA, NEAT, evolutionary computation, multimodal AI, OCR

Application Engineering: React, Next.js, Node.js, SwiftUI, REST APIs, PostgreSQL, Redis, Oracle Database, Auth0, Playwright, Konva

Cloud / DevOps: AWS S3, Docker, GitHub Actions, Git, CMake, Linux, CI/CD, Husky, ESLint, Prettier

Education & Recognition

Gannon University

2025

B.S. Computer Systems Engineering

Erie, PA

- Peer-reviewed publication with the American Society for Engineering Education (ASEE); research presentation at Sigma Xi.